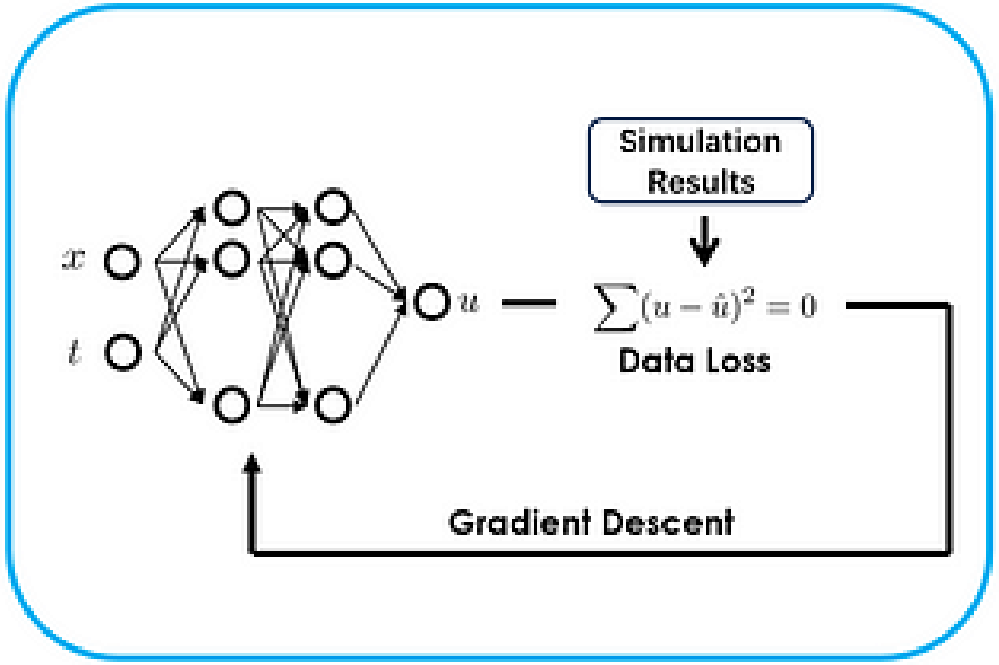
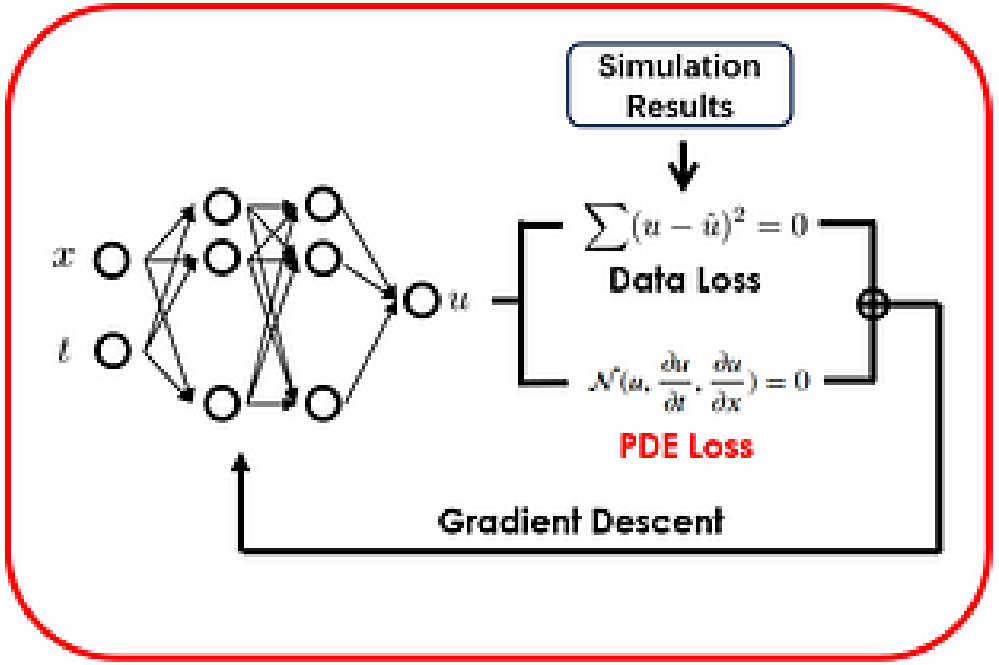
Lecture 20: PINNs Applications, DeepONets and Neural Operators

Sergei V. Kalinin

Back to functions from data



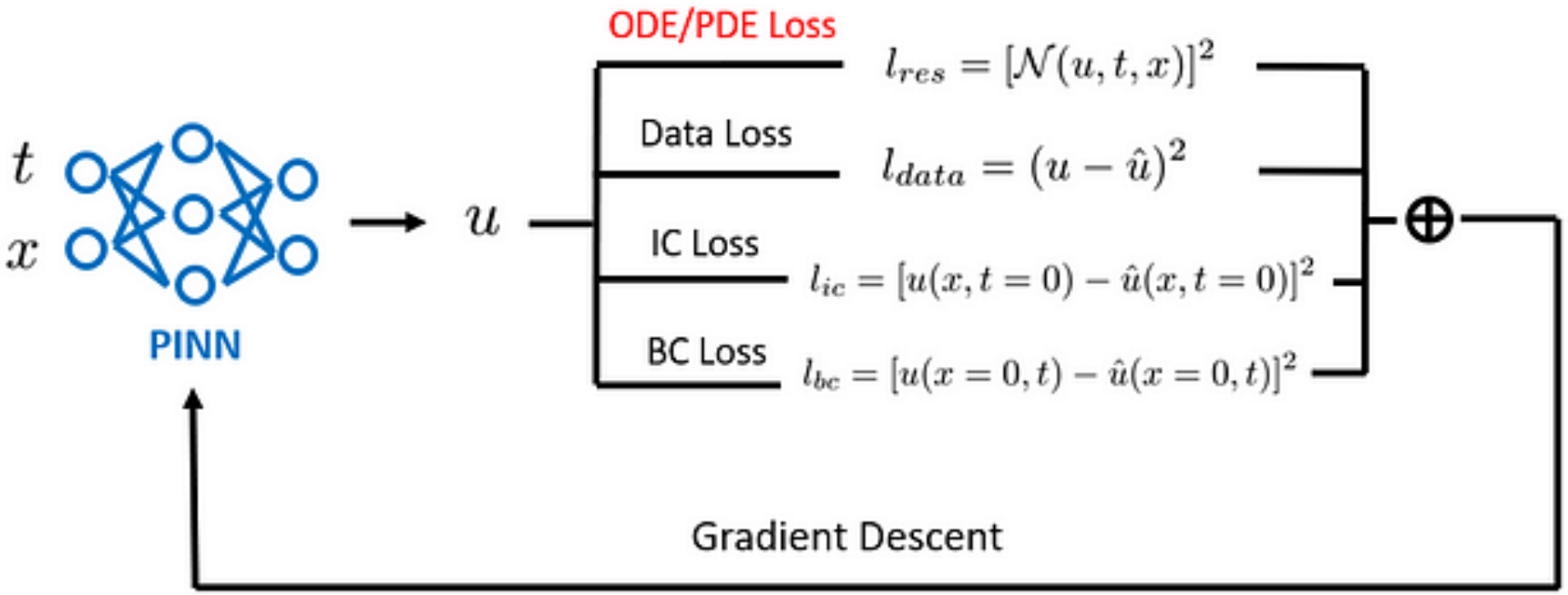
Tranditional NN



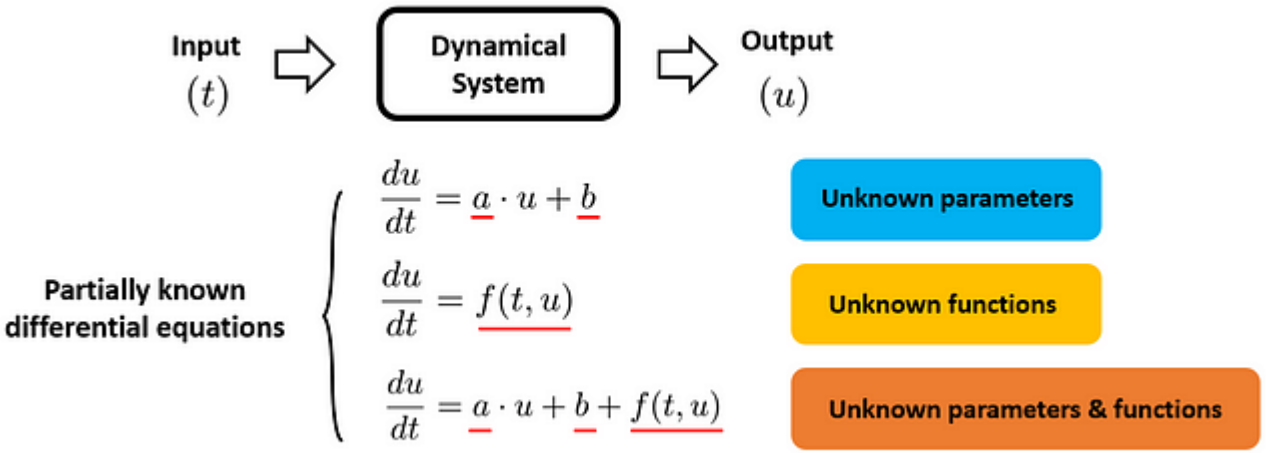
Physics-informed NN

https://towardsdatascience.com/physics-informed-neural-networks-an-application-centric-guide-dc1013526b02

PINN Problems



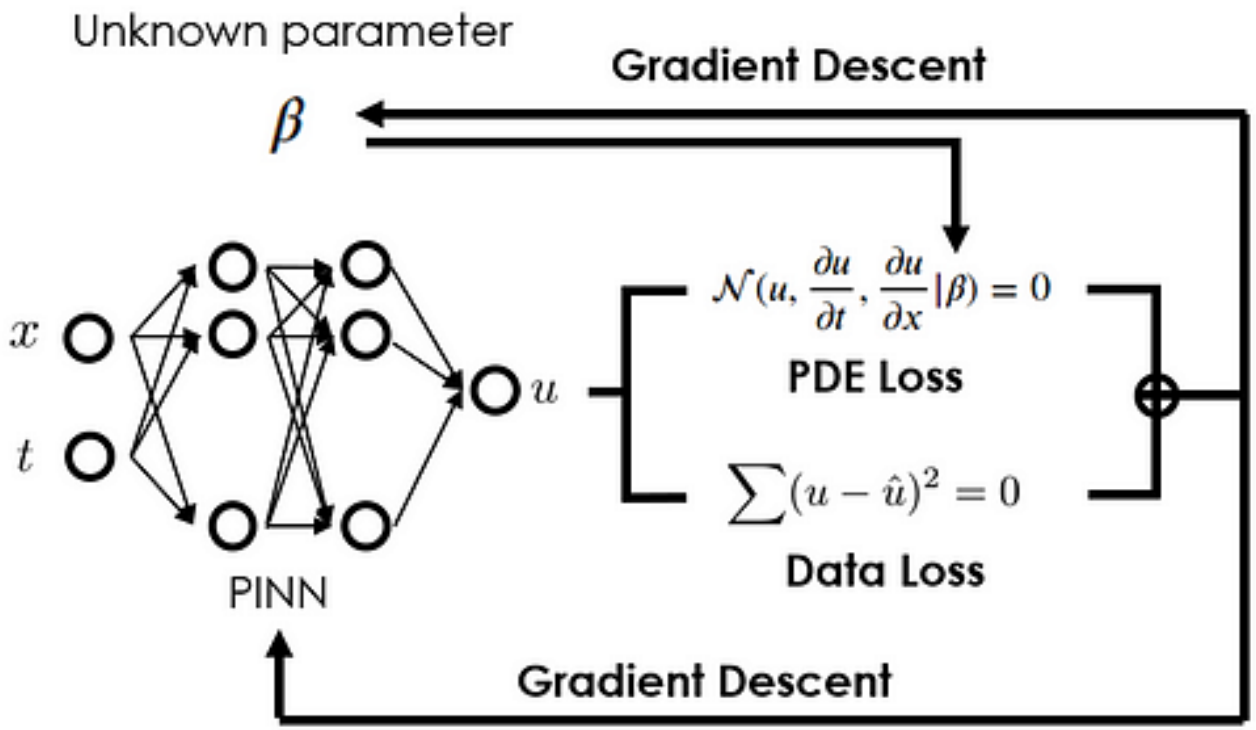
PINN Problems



- The **parameters** of the differential equation are unknown. For example, the governing equations of fluid dynamics are well-established, but the coefficients are highly uncertain.
- The **functional forms** of the differential equations are unknown. For instance, in chemical engineering, the exact functional form of the rate equations may not be fully understood due to the uncertainties in rate-determining steps and reaction pathways.
- Both **functional forms** and **parameters** are unknown. Example is battery state modeling, where the commonly used equivalent circuit model only partially captures the current-voltage relationship (the functional form of the missing physics is therefore unknown). The model itself contains unknown parameters (i.e., resistance and capacitance values).

[Discovering Differential Equations with Physics-Informed Neural Networks and Symbolic Regression | by Shuai Guo | Towards Data Science]

Parameter Estimation



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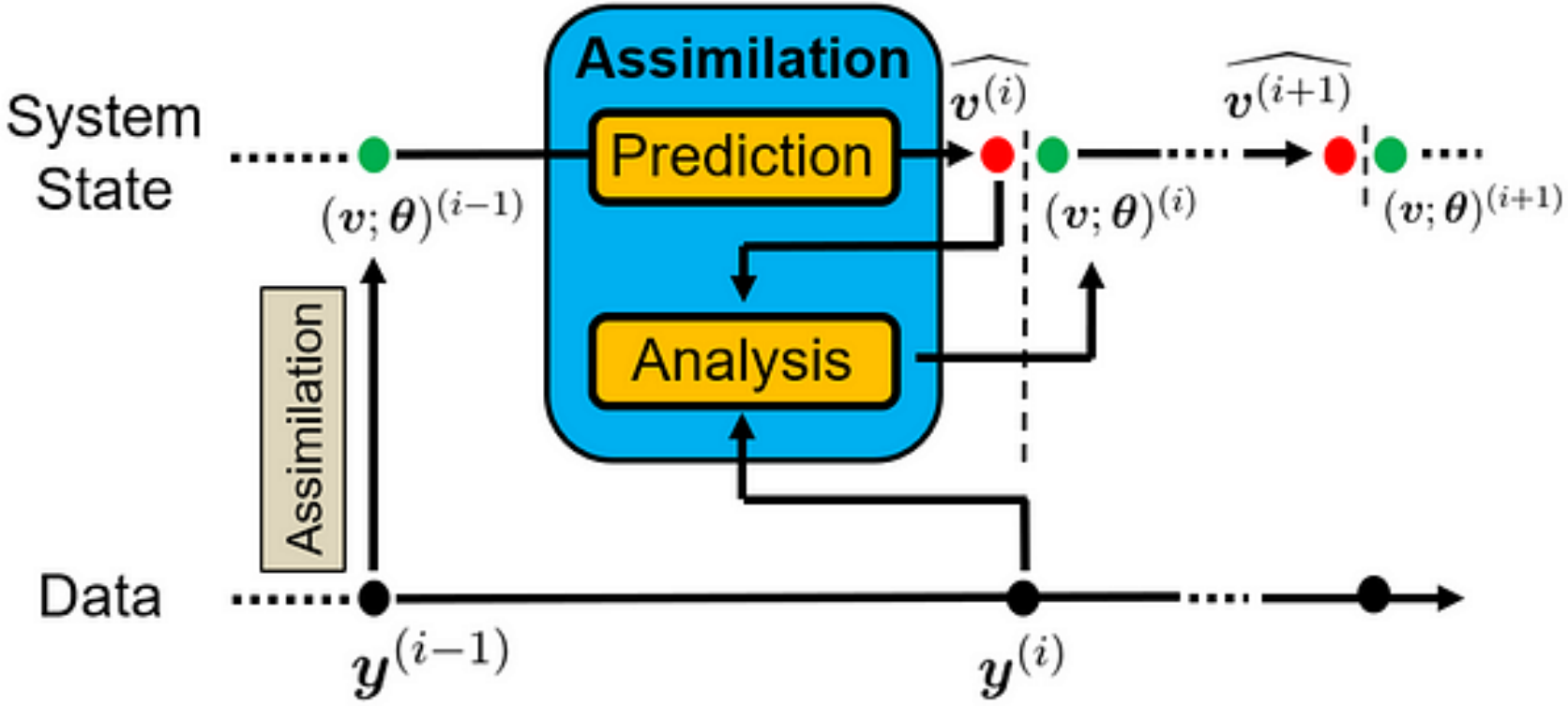
Data Assimilation



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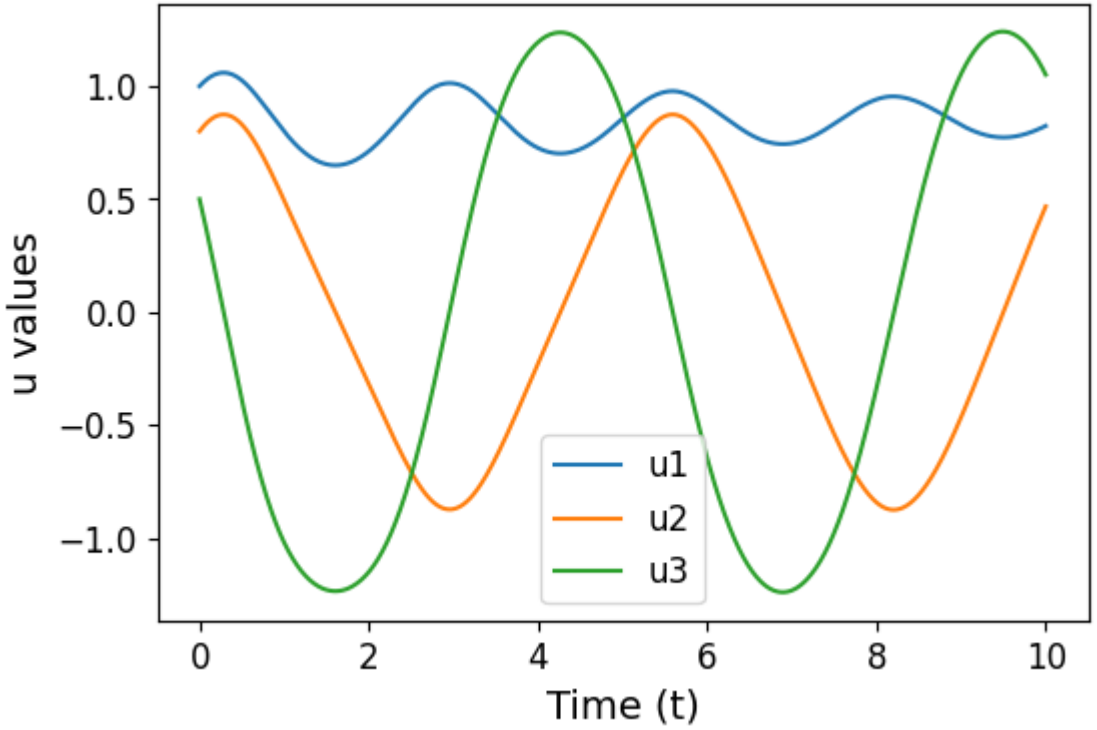
Craichnan-Orzag Equations

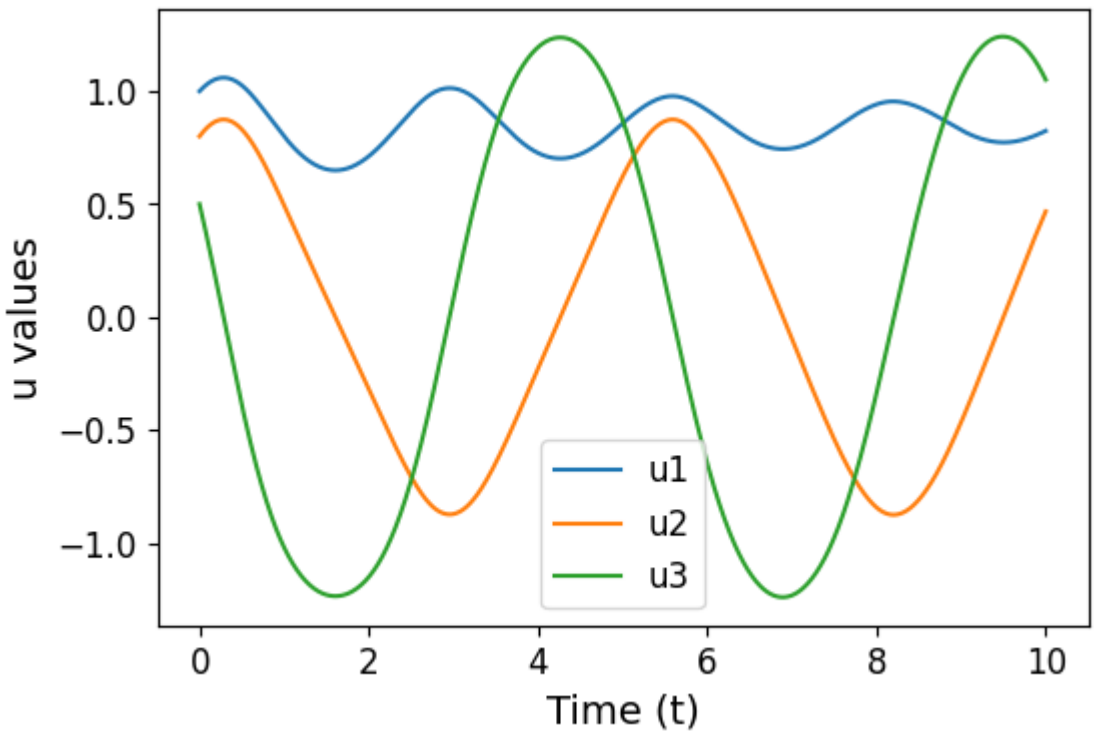
$$\frac{du_1}{dt} = e^{-t/10} u_2 u_3$$
$$\frac{du_2}{dt} = u_1 u_3$$
$$\frac{du_3}{dt} = -2 u_1 u_2$$



$u_1(0)=1$, $u_2(0)=0.8$, $u_3(0)=0.5$

https://medium.com/towards-data-science/discovering-differential-equations-with-physics-informed-neural-networks-and-symbolic-regression-c28d279c0b4d

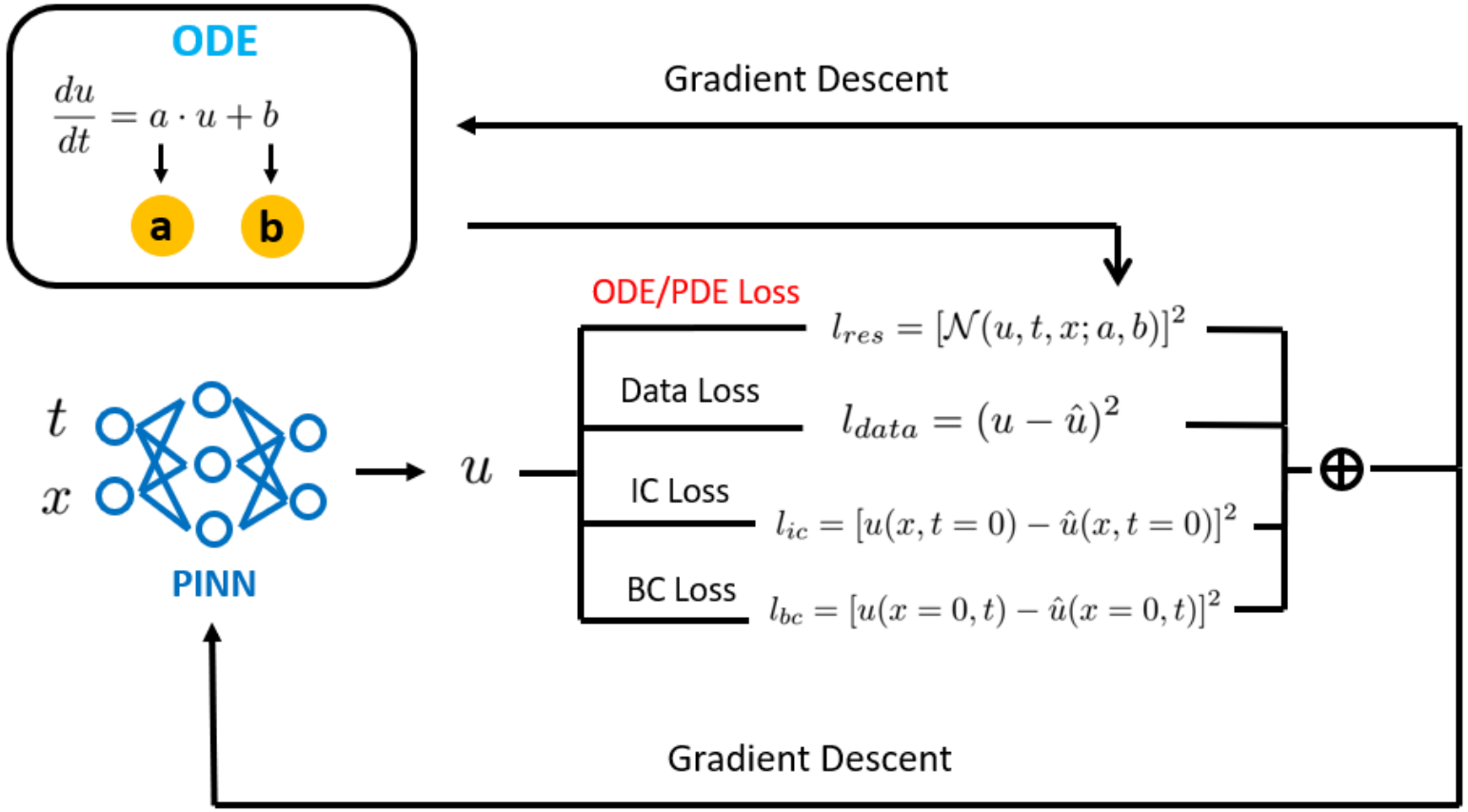
System Identification



$$\frac{du_1}{dt} = f_1(t, u_1, u_2, u_3)$$
$$\frac{du_2}{dt} = f_2(t, u_1, u_2, u_3)$$
$$\frac{du_3}{dt} = au_1u_2 + b$$

Here, f_1 and f_2 denote the unknown functions, and a and b are the unknown parameters. **Our objective is to calibrate the values of a and b, as well as estimate the analytical functional form of f_1 and f_2.**

https://medium.com/towards-data-science/discovering-differential-equations-with-physics-informed-neural-networks-and-symbolic-regression-c28d279c0b4d



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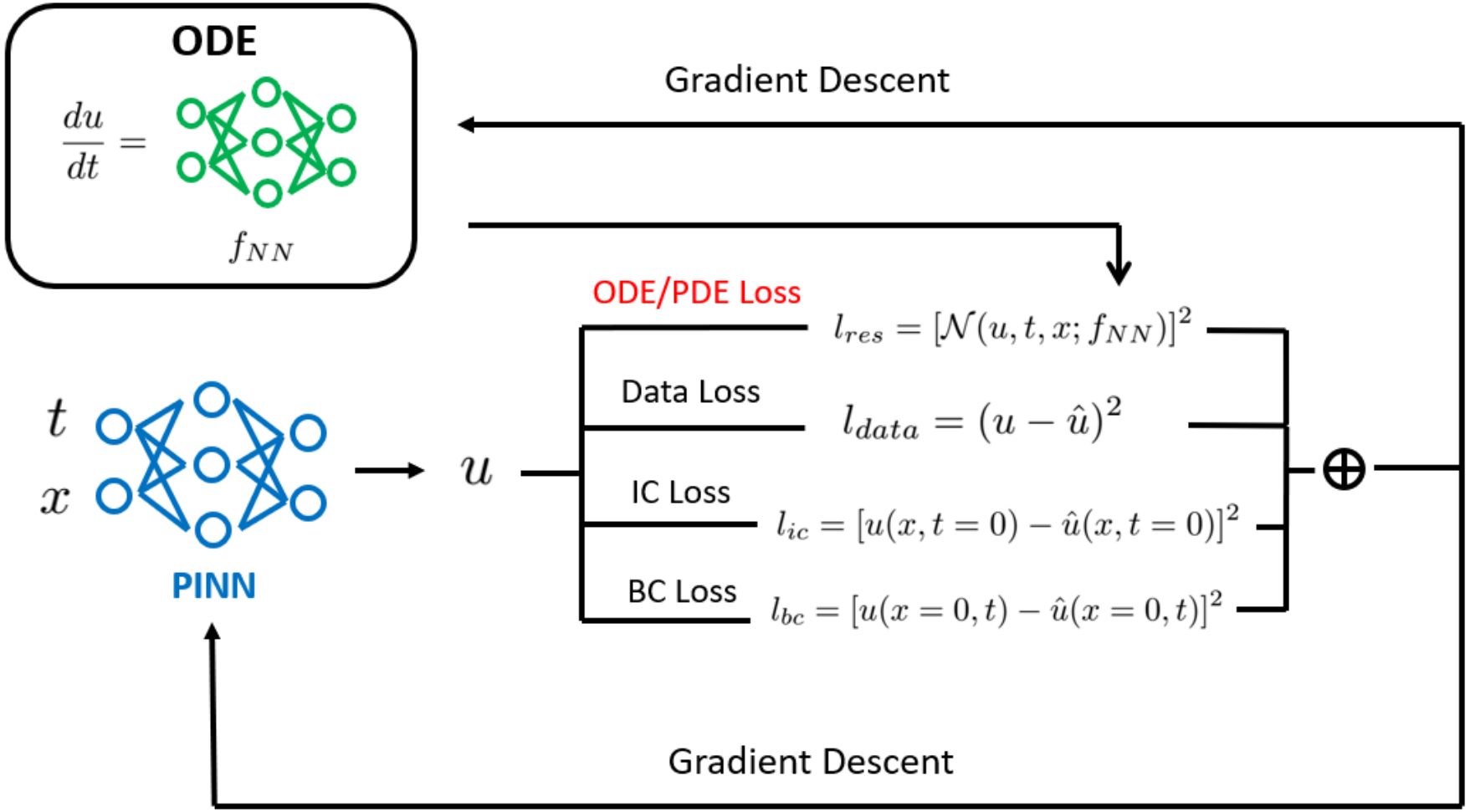

https://medium.com/towards-data-science/discovering-differential-equations-with-physics-informed-neural-networks-and-symbolic-regression-c28d279c0b4d

$$\frac{du_1}{dt} = f_1(t, u_1, u_2, u_3)$$
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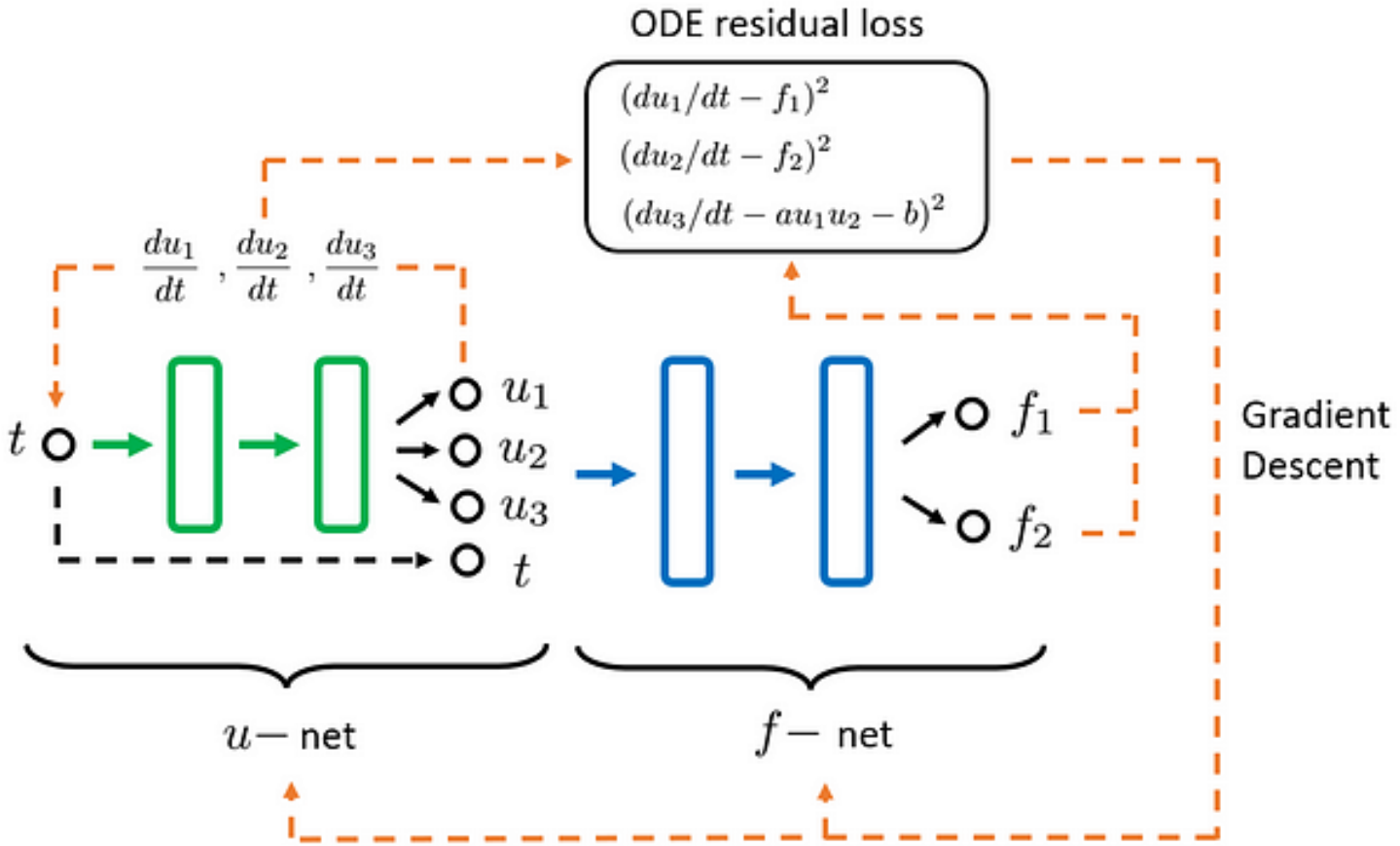

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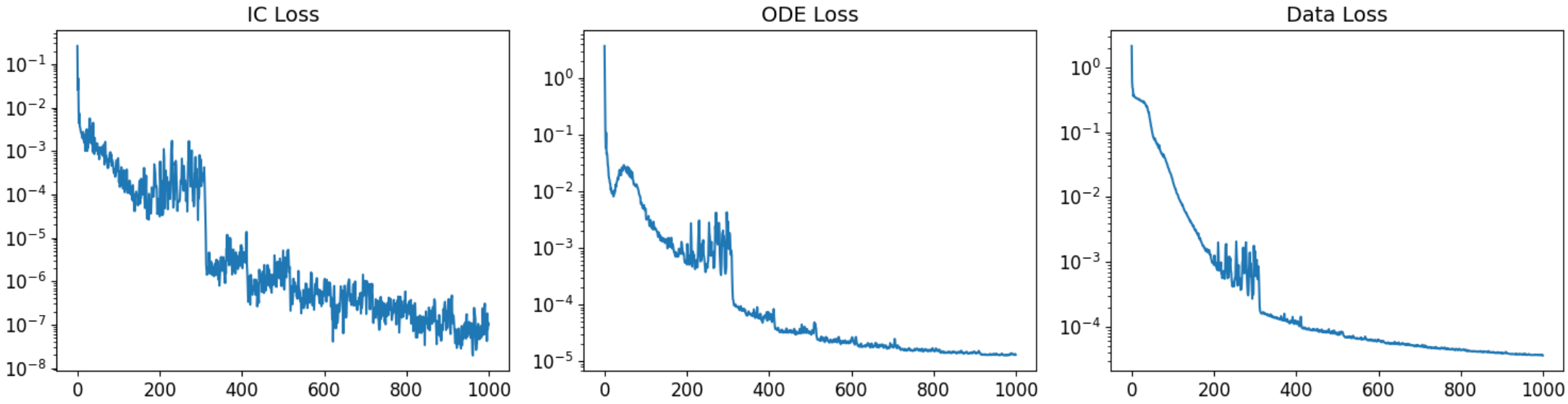
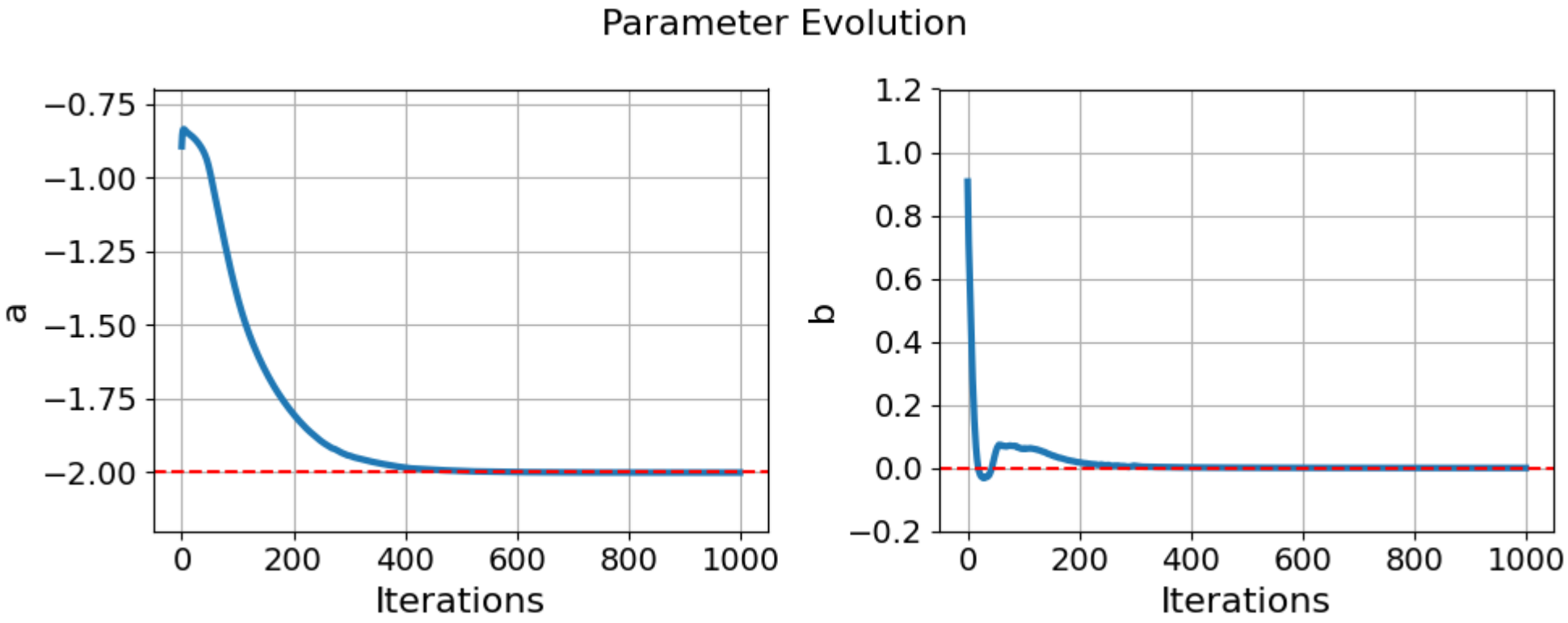
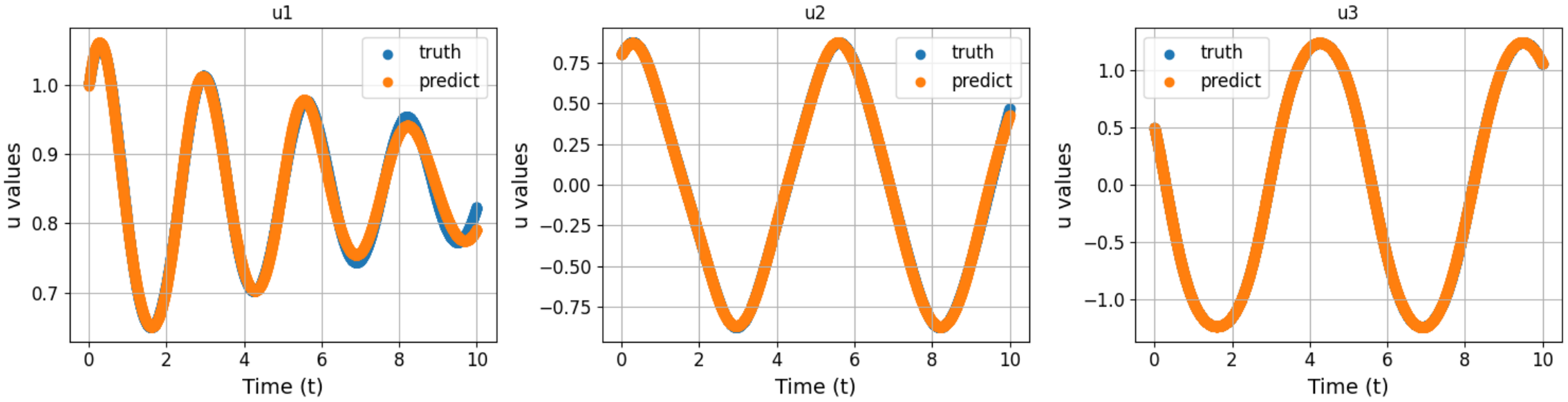
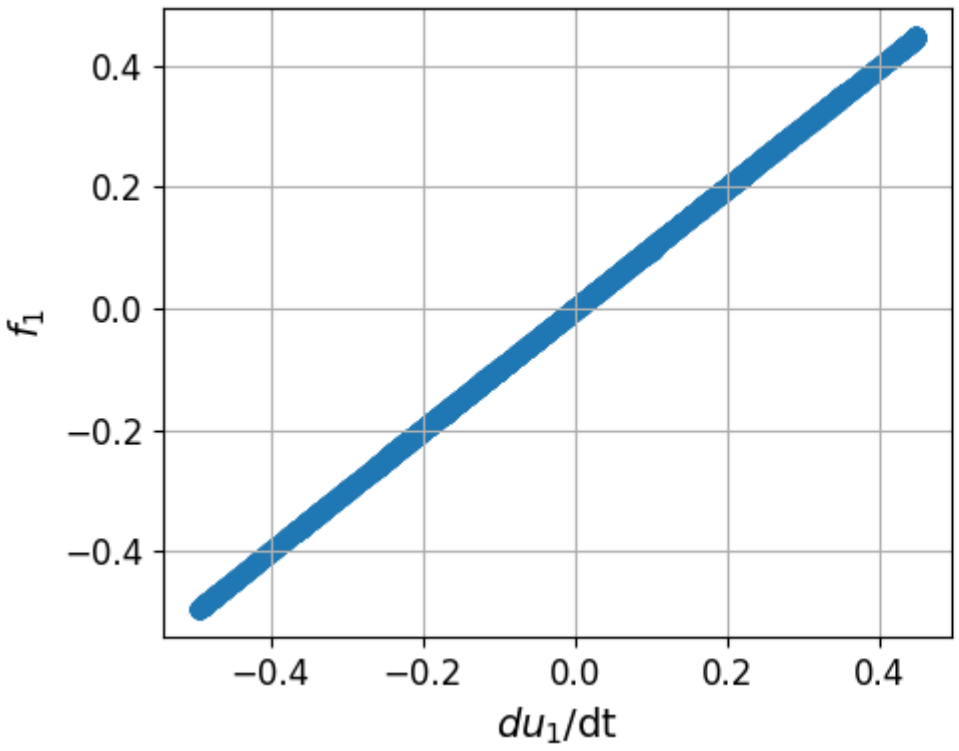
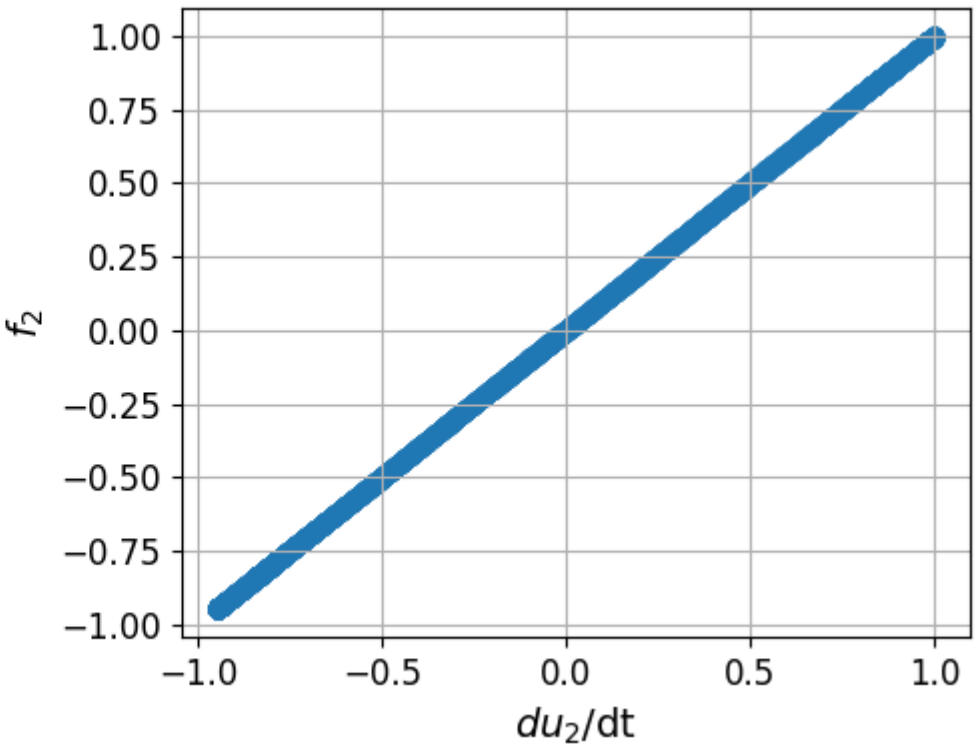

IC Loss
ODE Loss
Data Loss
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{0}
0
200
400
600
800
1000


Parameter Evolution
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
a
0
200
400
600
800
1000
Iterations
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
b
0
200
400
600
800
1000
Iterations


u1
truth
predict
u values
Time (t)
u2
truth
predict
u values
Time (t)
u3
truth
predict
u values
Time (t)


0.4
0.2
0.0
−0.2
−0.4
f_1
−0.4
−0.2
0.0
0.2
0.4
du_1/dt


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
f_2
−1.0
−0.5
0.0
0.5
1.0
du_2/dt

- Compared to the traditional numerical simulation approaches, PINNs are mesh-free. This enables PINNs to easily handle complex simulation domains, reduce manual effort in simulations, and potentially improve computational efficiency.

- Thanks to the universal approximation capability, PINNs are well-suited for modeling complex and nonlinear systems where traditional methods may struggle.

- Compared to the traditional paradigm of supervised learning, PINNs are in principle data-free, i.e., their training does not require collecting input-output pairs, i.e., (t, x, y) -(u, p, T), but entirely based on fulfilling the governing differential equations. However, if scarce data is available, PINNs can also effectively assimilate that.

https://towardsdatascience.com/physics-informed-neural-networks-an-application-centric-guide-dc1013526b02

- Heat equation:

https://inductiva.ai/blog/article/heat-1-an-introduction

https://inductiva.ai/blog/article/heat-2-pinn

https://inductiva.ai/blog/article/heat-3-neurosolver

Helmholz equation

https://towardsdatascience.com/improving-pinns-through-adaptive-loss-balancing-55662759e701

https://towardsdatascience.com/10-useful-hints-and-tricks-for-improving-pinns-1a5dd7b86001

https://towardsdatascience.com/unraveling-the-design-pattern-of-physics-informed-neural-networks-series-01-8190df459527

https://towardsdatascience.com/building-an-expert-gpt-in-physics-informed-neural-networks-with-gpts-75ebf6925966

Green's Functions

A **Green's function** is a fundamental solution used to solve **linear differential equations** with boundary conditions. It represents the **response of a system to a point source**, similar to how the **impulse response** works in signal processing.

Mathematically, for a **linear differential operator** L, the Green's function $G(x,x')$ satisfies:

$$LG(x,x')=\delta(x-x'),$$

where $\delta(x-x')$ is the **Dirac delta function**, representing a **unit impulse** at x'.

$G(x,x')$ is known, it can be used to **solve inhomogeneous differential equations** $Lu(x)=f(x)$ by integrating:

$$u(x)=\int G(x,x')f(x')dx'u(x)$$

Thus, Green's functions allow us to express the solution as a **superposition of responses to point sources**.

Green's Function in Electrostatics

Green's Function:

$$L \circ u(x) = f(x)$$

$$L \circ G(x,s) = \delta(x-s)$$

$$u(x) = \int G(x,s) f(s) ds$$

An Example:

$$\nabla^2 \varphi(\vec{r}) = -\frac{\rho(\vec{r})}{\varepsilon_0}$$

$$G(\vec{r},\vec{s}) = \frac{1}{4\pi\varepsilon_0} \frac{1}{|\vec{r}-\vec{s}|}$$

$$\varphi(\vec{r}) = \int G(\vec{r},\vec{s}) \rho(\vec{s}) d^3 s$$

https://galileo-unbound.blog/tag/greens-function/

Green's Functions depend on D and Domains

Table 10.1 Fundamental Green's Functions[a]

	Laplace ∇^2	Helmholtz[b] $\nabla^2 + k^2$	Modified Helmholtz[c] $\nabla^2 - k^2$
1-D	$\frac{1}{2}\lvert x_1 - x_2\rvert$	$-\frac{i}{2k}\exp(ik\lvert x_1 - x_2\rvert)$	$-\frac{1}{2k}\exp(-k\lvert x_1 - x_2\rvert)$
2-D	$\frac{1}{2\pi}\ln\lvert\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2\rvert$	$-\frac{i}{4}H_0^{(1)}(k\lvert\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2\rvert)$	$-\frac{1}{2\pi}K_0(k\lvert\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2\rvert)$
3-D	$-\frac{1}{4\pi}\frac{1}{\lvert\mathbf{r}_1 - \mathbf{r}_2\rvert}$	$-\frac{\exp(ik\lvert\mathbf{r}_1 - \mathbf{r}_2\rvert)}{4\pi\lvert\mathbf{r}_1 - \mathbf{r}_2\rvert}$	$-\frac{\exp(-k\lvert\mathbf{r}_1 - \mathbf{r}_2\rvert)}{4\pi\lvert\mathbf{r}_1 - \mathbf{r}_2\rvert}$

General Features of Green's Functions

1 Linearity – The differential equation must be linear since Green's functions rely on the superposition principle.

2 Specified Differential Operator – The equation must be in the form $Lu(x)=f(x)$ where LL is a known linear operator.

3 Specified Domain – The Green's function depends on the shape and size of the domain (e.g., infinite space, bounded region, irregular geometries).

4 Specified Boundary Conditions – The solution is uniquely determined only when boundary conditions (Dirichlet, Neumann, or mixed) are clearly defined.

5 Existence of Fundamental Solution – The Green's function must satisfy $LG(x,x')=\delta(x-x')$ and should be well-defined in the given domain.

6 Integral Representation of the Solution – The final solution must be expressible as an integral of the Green's function weighted by the source term

7 Solvability – In simple cases, Green's functions are found analytically; for complex domains, numerical techniques (e.g., finite elements, boundary elements) may be required.

Neural Operators

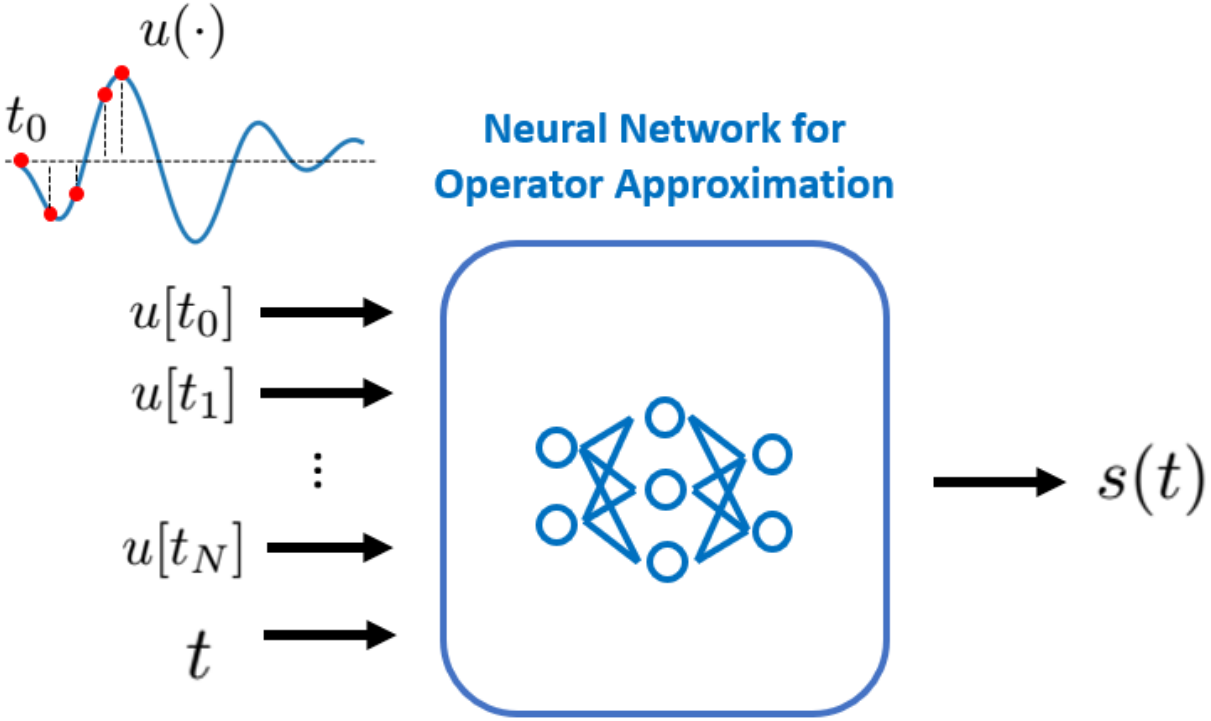
- **Nonlinear PDEs** – Green's functions require linearity

- **Variable Boundary Conditions** – Green's functions require a new derivation for each set of conditions

- **Irregular and Complex Geometries** – Green's functions are often limited to simple shapes

- **High-Dimensional Problems** – Green's functions are difficult to compute for **3D and high-dimensional PDEs**

- **Hybrid Data-Driven and Physics-Based Learning** – Green's functions need explicit physics



https://towardsdatascience.com/solving-inverse-problems-with-physics-informed-deeponet-a-practical-guide-with-code-implementation-27795eb4f502/

Universal Approximation Theorem

Theorem 1 (Universal Approximation Theorem for Operator). *Suppose that σ is a continuous non-polynomial function, X is a Banach Space, $K_1 \subset X$, $K_2 \subset \mathbb{R}^d$ are two compact sets in X and $\mathbb{R}^d$, respectively, V is a compact set in $C(K_1)$, G is a nonlinear continuous operator, which maps V into $C(K_2)$. Then for any $\epsilon > 0$, there are positive integers n, p, m, constants $c_i^k, \xi_{ij}^k, \theta_i^k, \zeta_k \in \mathbb{R}$, $w_k \in \mathbb{R}^d$, $x_j \in K_1$, $i = 1, \ldots, n$, $k = 1, \ldots, p$, $j = 1, \ldots, m$, such that*

$$\left| G(u)(y) - \sum_{k=1}^{p} \underbrace{\sum_{i=1}^{n} c_i^k \sigma \left(\sum_{j=1}^{m} \xi_{ij}^k u(x_j) + \theta_i^k \right)}_{branch} \underbrace{\sigma(w_k \cdot y + \zeta_k)}_{trunk} \right| < \epsilon \tag{1}$$

holds for all $u \in V$ and $y \in K_2$.

arXiv > cs > arXiv:1910.03193

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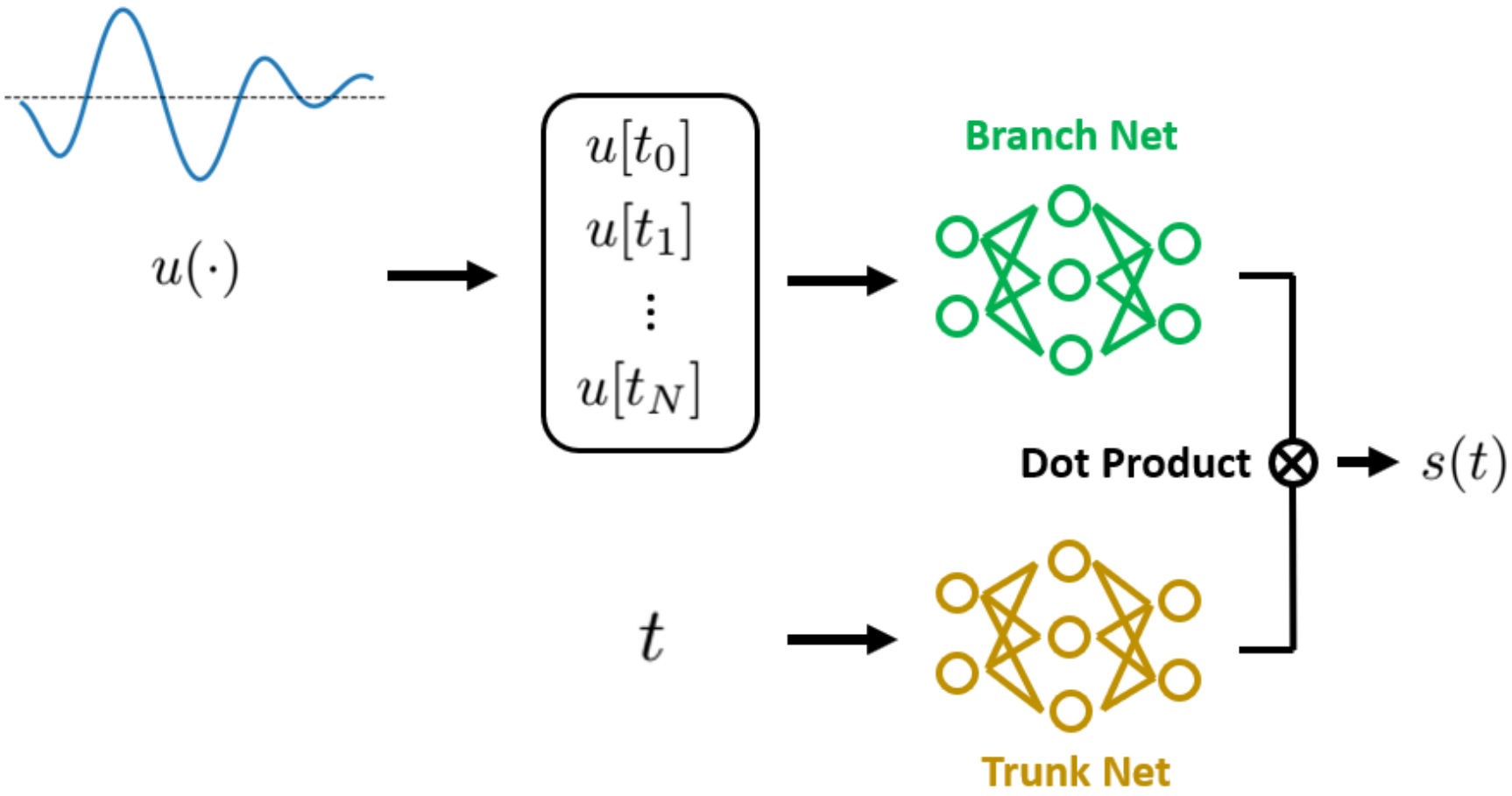
Computer Science > Machine Learning

[Submitted on 8 Oct 2019 (v1), last revised 15 Apr 2020 (this version, v3)]

DeepONet: Learning nonlinear operators for identifying differential equations based on the universal approximation theorem of operators

Lu Lu, Pengzhan Jin, George Em Karniadakis

DeepONet



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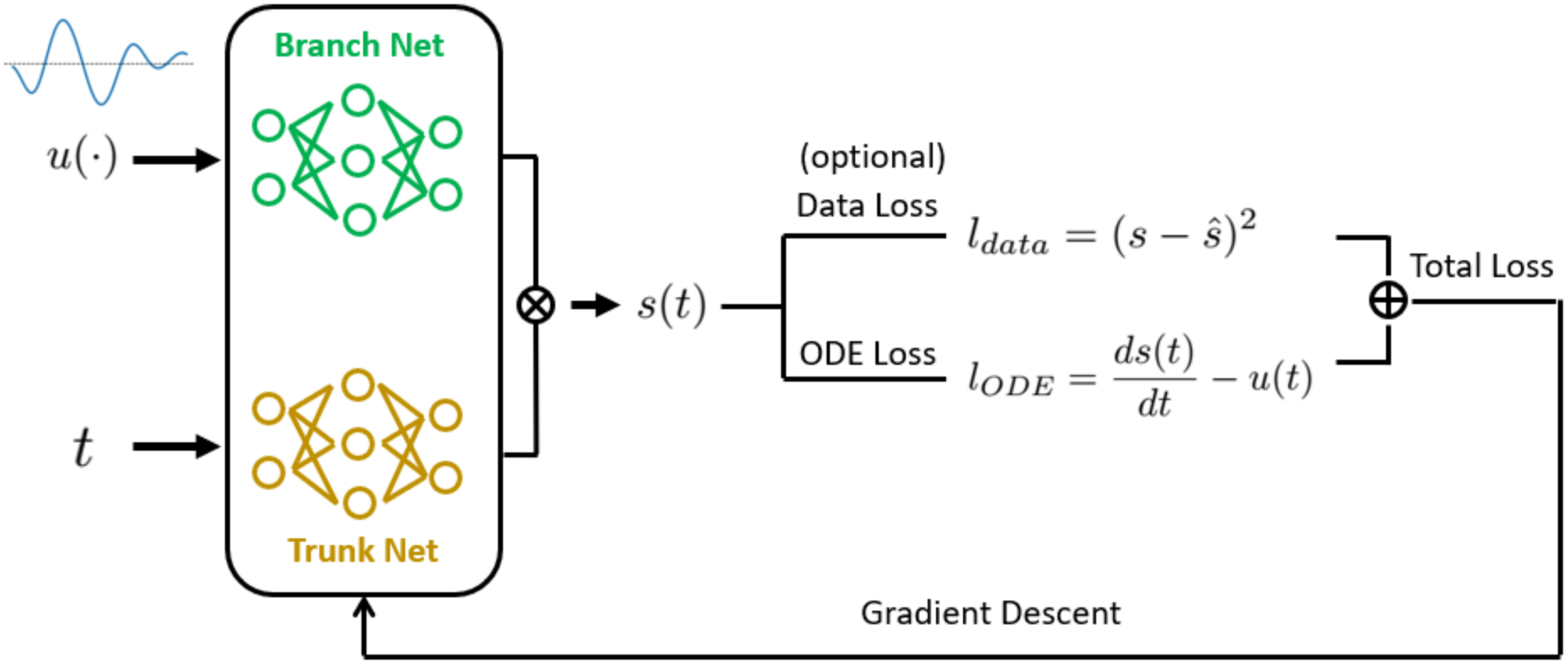
Physics-Informed DeepONet



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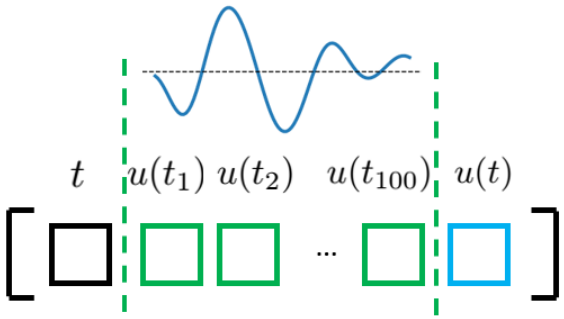
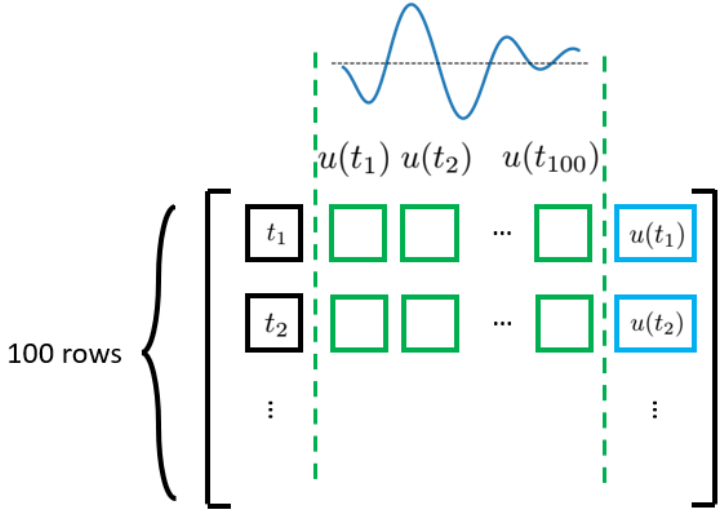
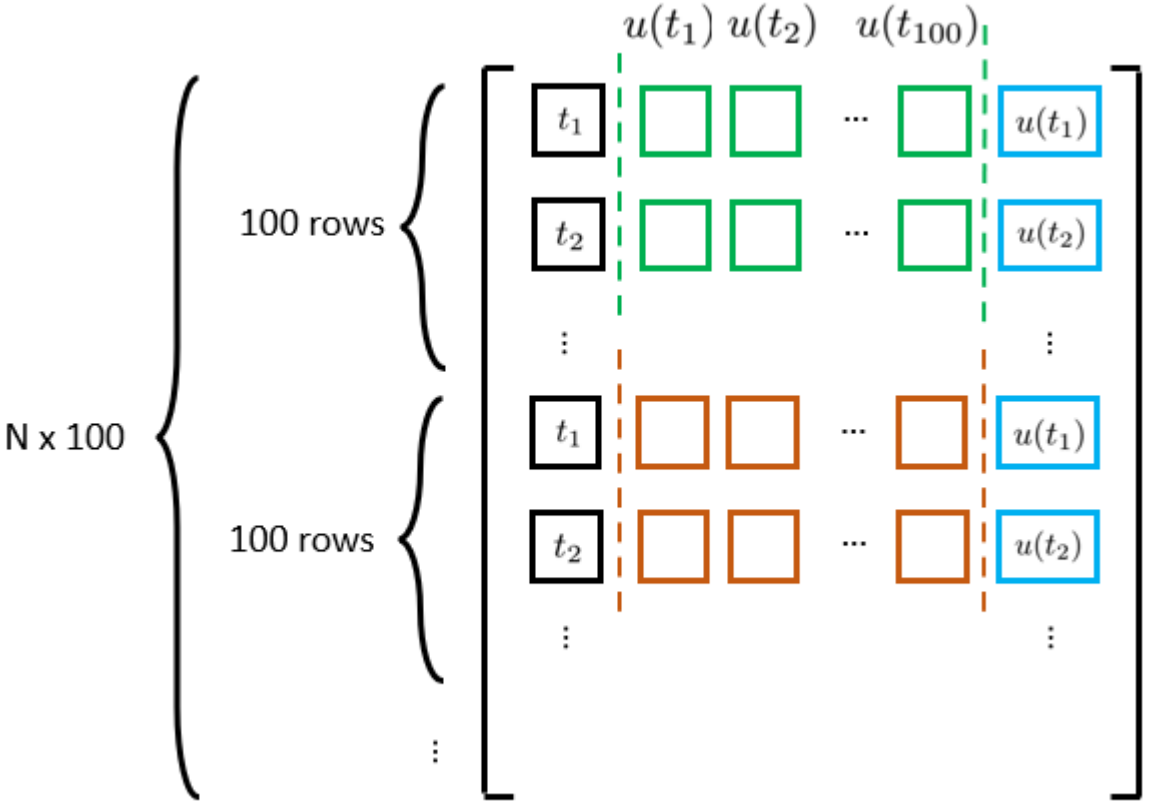
DeepONet Training







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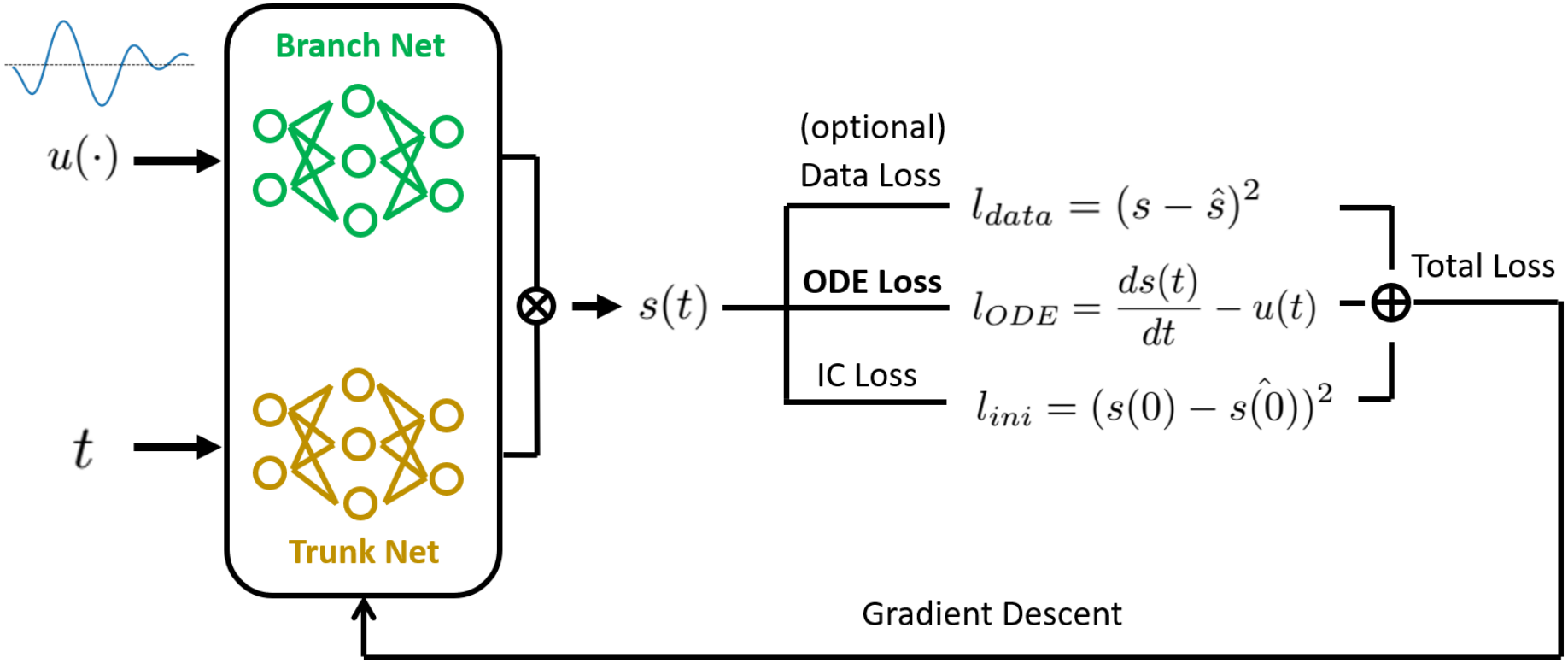
Physics-Informed DeepONet



https://towardsdatascience.com/solving-inverse-problems-with-physics-informed-deeponet-a-practical-guide-with-code-implementation-27795eb4f502/

Physics-Informed DeepONet Applications

Parameter discovery

$$\frac{ds(t)}{dt} = a \cdot u(t) + b, \ t \in [0, 1]$$

s(0) = 0, a and b are **unknowns**. Our objective here is to estimate the values of *a* and *b*, given the observed u(·) and s(·) profiles.

Function discovery

$$\frac{ds(t)}{dt} = 0.5 \cdot u(t) + 0.5, \ t \in [0, 1]$$

initial condition s(0) = 0, a=0.5, b=0.5. We aim to learn u(t)

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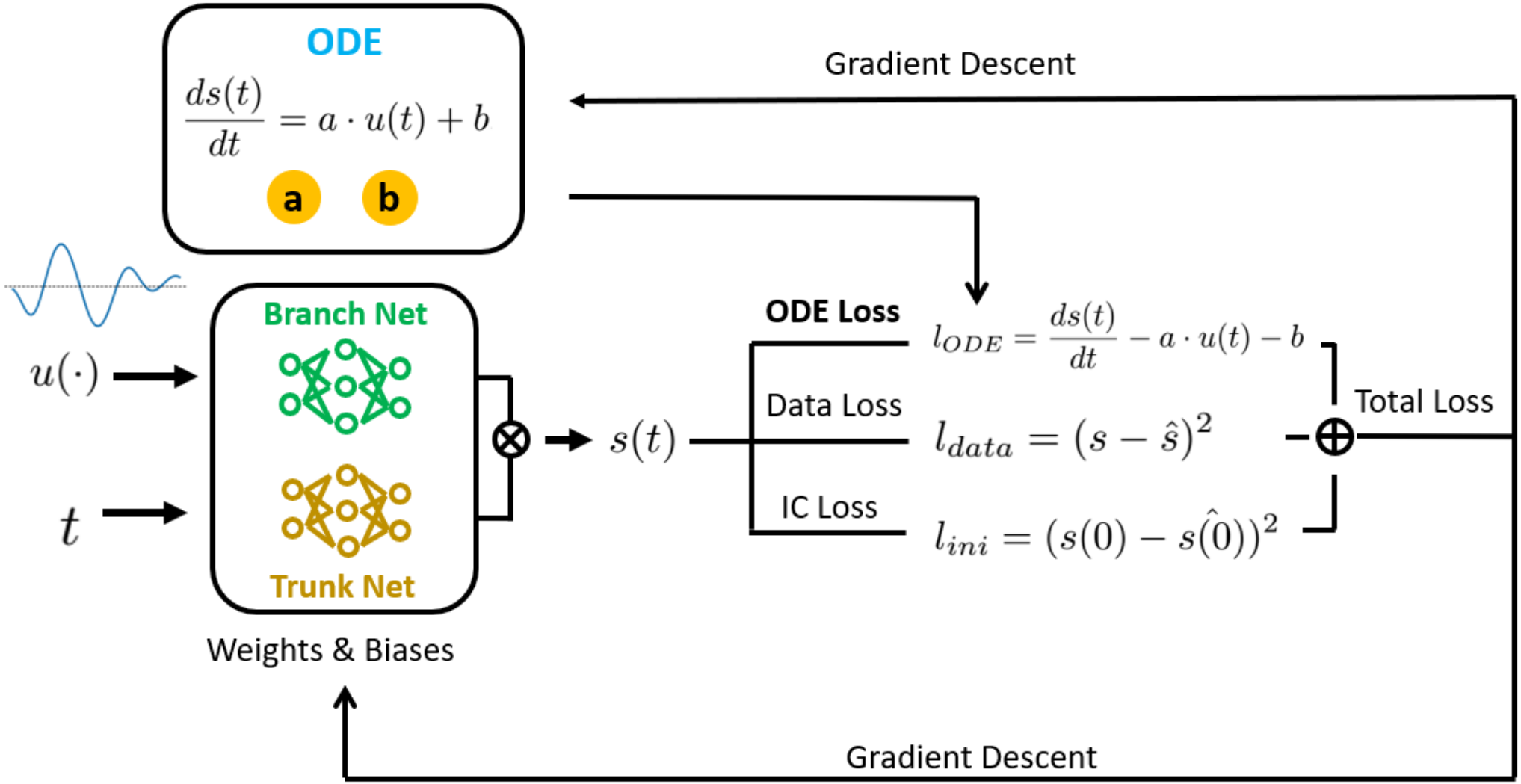
Physics-Informed DeepONet Applications



https://towardsdatascience.com/solving-inverse-problems-with-physics-informed-deeponet-a-practical-guide-with-code-implementation-27795eb4f502/

Green's Functions vs. PI-DeepONets

• **Analytical Exact Solutions** – Green's functions provide closed-form solutions, while DeepONets are approximations.
• **Infinite Domains** – Green's functions can naturally handle infinite space problems, while DeepONets require **training on finite domains**.
• **Singular Solutions** – Green's functions handle **point singularities** well, while DeepONets may struggle with sharp features.
• **Universal Applicability** – Green's functions work for any system that satisfies the linear PDE, whereas DeepONets must be trained for each problem.
• **Lack of Data Dependency** – Green's functions can be derived directly from physics, while DeepONets require **training data or simulations**.

- **Linearity -** Green's functions require linearity, while DeepONets can approximate solutions for nonlinear equations.
- **Variable Boundary Conditions** – Green's functions require a new derivation for each set of conditions, while DeepONets can generalize across different boundaries.
- **Irregular and Complex Geometries** – Green's functions are often limited to simple shapes, whereas DeepONets can learn solutions in **arbitrary domains**.
- **High-Dimensional Problems** – Green's functions are difficult to compute for **3D and high-dimensional PDEs**, while DeepONets scale more efficiently.
- **Hybrid Data-Driven and Physics-Based Learning** – Green's functions need explicit physics, but DeepONets can **combine physics with real-world data**.

Neural Operators

1 **Nonlinear PDEs –** Green's functions rely on **linearity**, and DeepONets struggle with **generalizing across nonlinear interactions**. What about **nonlinear function mappings** ?

2 **New Domains & Geometries –** Green's functions are **derived for specific shapes** (e.g., rectangles, spheres), and DeepONets are trained on **fixed-sized inputs**. What about **arbitrary domains** (without retraining)?

3 **Changing Boundary Conditions –** Green's functions require **re-derivation** for each new boundary condition, and DeepONets need **new training**. What about new **boundary variations?**

4 **High-Dimensional PDEs –** Green's functions involve **expensive integration**, and DeepONets require **training on large datasets**. How can we scale to **high-dimensional systems?**

5 **Data-Scarce Problems –** DeepONets need **large training datasets**, Can we learning continuous function mappings with fewer datapoints?

6 **Multiscale & Complex Physics –** Problems like **turbulence, climate modeling, and coupled physical systems** require models that generalize **across scales and conditions**, which Green's functions and DeepONets struggle with.

Neural Operators

arXiv > cs > arXiv:2108.08481

Computer Science > Machine Learning

[Submitted on 19 Aug 2021 (v1), last revised 2 May 2024 (this version, v6)]

Neural Operator: Learning Maps Between Function Spaces

Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, Anima Anandkumar

Property \ Model	NNs	DeepONets	Interpolation	Neural Operators
Discretization Invariance	✗	✗	✓	✓
Is the output a function?	✗	✓	✓	✓
Can query the output at any point?	✗	✓	✓	✓
Can take the input at any point?	✗	✗	✓	✓
Universal Approximation	✗	✓	✗	✓

Table 1: Comparison of deep learning models. The first row indicates whether the model is discretization invariant. The second and third rows indicate whether the output and input are a functions. The fourth row indicates whether the model class is a universal approximator of operators. Neural Operators are discretization invariant deep learning methods that output functions and can approximate any operator.

Neural Operators generalize across different geometries and boundary conditions without retraining - a significant advantage for applications like climate modeling and fluid dynamics

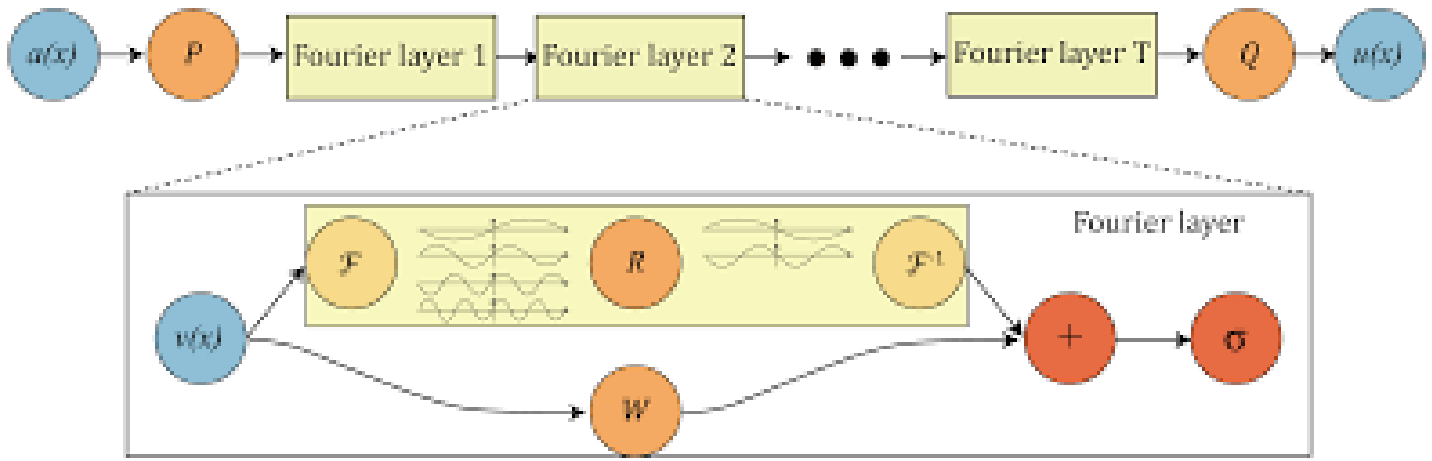
Neural Operators vs. PINNs and Solvers

Feature	Neural Operators	PINNs	Classic Solvers
What is learned?	Function-to-function mapping	Single PDE solution	Single PDE solution
Training	Data-driven	Physics-constrained	Not needed
Computational Cost	Expensive to train, fast inference	Moderate	Expensive
Generalization	Can generalize across PDE families	Solves one PDE at a time	One equation per simulation
Best for	High-dimensional PDEs, complex geometries	Data-sparse problems, physics-driven tasks	Benchmarking, high-precision cases

Neural Operators: A Peek Below the Hood

Neural Operators aim to approximate **operator mappings** between function spaces. Given an input function f(x) defined over a domain Ω, a Neural Operator learns a mapping: G:f↦u where u(x) is the solution of a differential equation governed by f(x), and G represents the **solution operator**.

Let D be a space of functions defined on a domain Ω, and let $G{:}D{\rightarrow}U$ be an operator mapping from the input function space D (e.g., forcing terms, initial conditions) to the output function space U (e.g., solution of a PDE). For a given function f, the corresponding solution is: $u{=}G(f)$. The goal of Neural Operators is to learn G from data.



Neural Operators: A Peek Below the Hood

Step 1: Lifting the Input to a High-Dimensional Space
Define a **lifting operator** PPP that embeds the input function into a higher-dimensional space:
$v_0(x)=P(f(x))$ where v_0 is an intermediate representation in an expanded feature space.

Step 2: Learning the Integral Kernel Operator
Instead of traditional convolutional layers, Neural Operators use a **global integral kernel** K_θ to model interactions between function values:

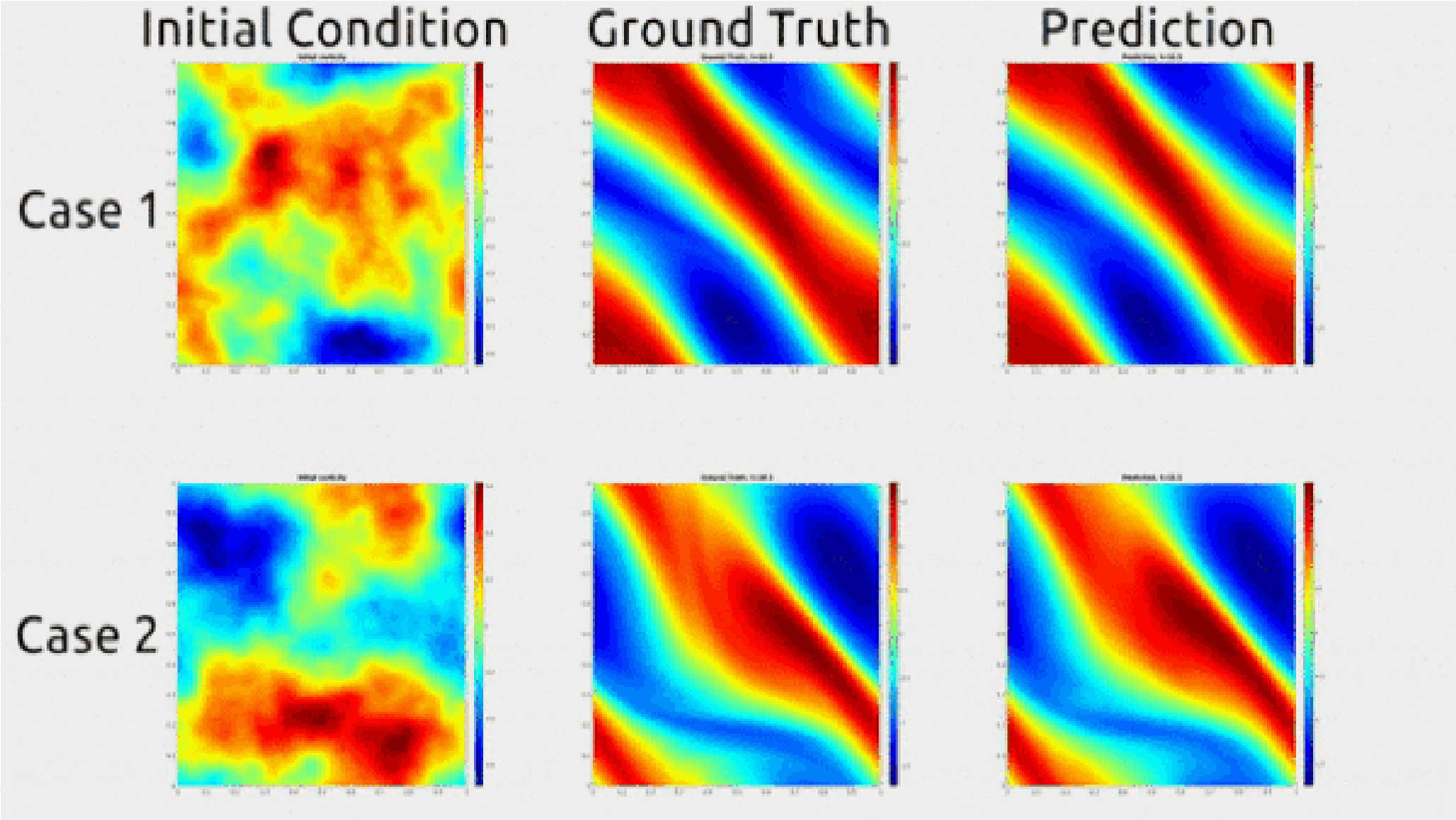
$$v_{k+1}(x)=\sigma\left(\int_\Omega K_\theta(x,x')v_k(x')\,dx'+Wv_k(x)\right)$$

where:
- $K_\theta(x,x')$ is the learned **integral kernel** that encodes relationships between points x and x'.
- W is a local transformation (e.g., pointwise linear operator).
- σ is a nonlinear activation function.

This equation defines how function values interact across the entire domain, rather than just locally.

Step 3: Projection Back to Output Space
After several layers of kernel updates, the final output is obtained by applying a **projection operator** Q:
$u(x)=Q(v_k(x))$
This ensures that the output function remains in the correct solution space.

Neural Operators Examples



https://zongyi-li.github.io/neural-operator/

Feature	Green's Functions	DeepONet	Neural Operators
Problem Type	Linear differential equations	Data-driven function approximation	Learning operators for PDEs
Boundary Conditions	Required, explicitly handled	Implicitly learned from data	Can incorporate, but not always explicitly defined
Domain Shape	Typically simple (e.g., rectangular, spherical)	Can handle irregular domains if trained on them	Works on general domains, flexible input structures
Input Data	Differential equation and source term	Function values at discrete points	Function values, can handle grid-based or irregular data
Solution Representation	Integral over Green's function	Neural network-based mapping	Maps functions to functions directly
Training Data Needed?	No, purely analytical	Yes, needs training data	Yes, needs training data for operator learning
Computational Cost	High for complex problems, needs explicit integral	Training is costly, but inference is fast	Training is costly, but efficient at inference
Adaptability to New Problems	Must derive new Green's function for each problem	Can generalize to similar problems if trained properly	Highly generalizable to different PDEs and domains
Memory Usage	Large for storing Green's functions in high dimensions	Moderate, depends on network size	More efficient for large-scale problems

Type of Error	Definition	Causes	How to Reduce
Approximation Error	Difference between best possible function in model class and the true function f∗f^*f∗	Limited model capacity, wrong architecture	Increase network size, change activation functions, improve architecture
Optimization Error	Difference between best possible function and what is learned during training	Poor optimizer, bad learning rate, local minima, bad initialization	Use Adam/SGD, adjust learning rate, batch norm, weight initialization
Generalization Error	Difference between training error and test error	Overfitting, small dataset, high variance	Regularization, more data, dropout, early stopping